Canonicalization Leakage: How Canonical Representatives Confound Supervised Learning under Group Symmetry

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Abstract

We identify *canonicalization leakage*, a failure mode in which models trained on canonical representatives under group symmetry exploit representation artifacts rather than learning invariant structure. When a dataset selects one element per G-orbit via a deterministic canonicalization procedure, the procedure imposes non-invariant distributional structure that sufficiently expressive learners can exploit. We formalize this phenomenon and propose CL-DIAG, a diagnostic protocol for datasets with known finite group symmetry. The mechanism accounts for earlier observations of SMILES sensitivity in molecular ML, which were documented empirically without a group-theoretic framing. We demonstrate CL-DIAG on circuit complexity prediction for 616,126 NPN classes of 5-input Boolean functions ($|G|$=7,680). An MLP reaches $r_s = 0.788$ on canonical inputs, yet 0% prediction consistency across NPN-equivalent inputs—dropping to $r_s = 0.254$ on non-canonical forms. Per-feature invariance testing yields an ordering: truth table bits (std=1.66) $>$ Fourier spectrum (0.97) $>$ invariant measures (0.00). Symmetry-consistent augmentation (7$\times$ NPN copies) recovers $r_s = 0.777$ on NPN-averaged evaluation, 14pp above the classical invariant-measure ceiling (0.635). A matched-volume control experiment separates group-consistent transformation from generic regularization: NPN augmentation (r_s=0.728) versus data duplication (0.172) at equal volume. CL-DIAG applies to any domain with canonical representatives under known group symmetry, including molecular ML, graph classification, and crystallography.

1 Introduction

Many supervised learning tasks possess group symmetry: a group G acts on the input space $\mathcal{X}$, and the target function f^* satisfies $f^*(g \cdot x) = f^*(x)$ for all $g \in G$. Datasets for such tasks often consist of *canonical representatives*—one element per G-orbit, selected by a deterministic canonicalization procedure. This is common: molecular property prediction uses canonical SMILES strings [Bjerrum, 2017], graph classification uses nauty canonical labelings [McKay and Piperno, 2014], crystallography uses Niggli-reduced unit cells [Niggli, 1928], and Boolean function databases store one NPN-canonical truth table per equivalence class [Knuth, 2015].

Canonicalization is typically treated as neutral preprocessing: since f^* is G-invariant, restricting to canonical representatives preserves the target while eliminating redundancy. We show this reasoning is incomplete. A model trained on canonical representatives can exploit *non-invariant features of the*

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canonical form itself, achieving high apparent performance without learning invariant structure. We call this **canonicalization leakage**.

Canonicalization operators $C : \mathcal{X} \to \mathcal{X}$ select a specific orbit representative by arbitrary convention (lexicographic ordering, atom numbering, variable indexing). These conventions impose distributional structure on the canonical dataset that is absent from the underlying orbits. Any sufficiently expressive learner can exploit this structure, producing models that succeed on canonical test data but fail on non-canonical inputs from the same orbits.

We formalize canonicalization leakage (Section 2) and propose CL-DIAG, a diagnostic protocol that detects, localizes, and quantifies it. We apply the protocol to circuit complexity prediction for 5-input Boolean functions under the NPN group ($|G| = 7{,}680$), where the target is exactly known for all $616{,}126$ equivalence classes [Knuth, 2015, Goucher, 2019] and classical invariants provide a known performance ceiling.

Independent observations of this phenomenon exist in molecular ML: canonical SMILES [Weininger, 1988] degrade performance relative to randomized SMILES [Bjerrum, 2017, Arús-Pous et al., 2019], and recent work has tested chemical LLM robustness to SMILES variation [Steshin, 2025]. These studies attributed the degradation to insufficient data diversity or representation fragility rather than to a structural property of canonicalization. Equivariant architectures [Cohen and Welling, 2016, Bronstein et al., 2021] can eliminate the problem by construction but do not explain why non-equivariant models trained on canonical data fail. We formalize the mechanism and provide a diagnostic protocol, signal decomposition, and matched-volume controls for symmetry-reduced settings.

Contributions.

1. **Formalization of canonicalization leakage** as a failure mode in symmetry-reduced supervised learning, with a signal taxonomy decomposing apparent performance into invariant signal ($S_{\text{inv}} + S_{\text{neural}}$) and leakage ($L_{\text{canon}}$).
2. **CL-DIAG**: a diagnostic protocol (train $\to$ invariance test $\to$ contamination decomposition $\to$ augmented training $\to$ causal control $\to$ signal decomposition) applicable to datasets with known finite group symmetry.
3. **Diagnosis on circuit complexity prediction**: NPN invariance testing reveals 0% prediction consistency; per-feature decomposition identifies a hierarchy from truth table bits (most contaminated) to invariant measures (zero leakage).
4. **Invariant signal recovery**: symmetry-consistent augmentation achieves $r_s = 0.777$, 14pp above the classical invariant-measure ceiling (0.635)—the network extracts NPN-invariant structure beyond the 7 handcrafted measures.
5. **Matched-volume controls** that isolate group-consistent transformation as the active mechanism over data volume and generic regularization.

2 Canonicalization Leakage: A General Framework

2.1 Formal Setup

Definition 1 (Symmetric Learning Problem)**.** *A* symmetric learning problem *is a tuple* $(\mathcal{X}, G, \varphi, f^*)$ *where* $\mathcal{X}$ *is the input space,* G *is a finite group acting on* $\mathcal{X}$*,* $\varphi : \mathcal{X} \to \mathbb{R}^d$ *is a feature map, and* $f^* : \mathcal{X} \to \mathbb{R}$ *is the target function satisfying* $f^*(g \cdot x) = f^*(x)$ *for all* $g \in G, x \in \mathcal{X}$*.*

Definition 2 (Canonicalization Operator)**.** *A* canonicalization operator $C : \mathcal{X} \to \mathcal{X}$ *selects exactly one representative per* G*-orbit:* $C(g \cdot x) = C(x)$ *for all* $g \in G$*, and* $C(x) \in [x]_G$ *where* $[x]_G = \{g \cdot x : g \in G\}$ *is the orbit of* x*.*

Definition 3 (Canonical Dataset)**.** *Given a set of* G*-orbits* $\mathcal{O} = \{[x_1]_G, \ldots, [x_N]_G\}$*, the* canonical dataset *is* $\mathcal{D}_C = \{(\varphi(C(x_i)),\ f^*(x_i))\}_{i=1}^N$*.*

The feature map φ decomposes into G-invariant and non-invariant components: $\varphi(x) = (\varphi_{\text{inv}}(x),\ \varphi_{\text{non}}(x))$, where $\varphi_{\text{inv}}(g \cdot x) = \varphi_{\text{inv}}(x)$ for all $g \in G$, and φ_{non} varies across the orbit.

Definition 4 (Canonicalization Leakage)**.** *Let $\hat{f}$ be a model trained on $\mathcal{D}_C$. The* canonicalization leakage *of $\hat{f}$ is $\delta(\hat{f}) = \mathit{Perf}(\hat{f}, \mathcal{D}_C) - \mathit{Perf}_G(\hat{f})$, where $\mathit{Perf}_G(\hat{f}) = \mathbb{E}_{x,g\sim \mathit{Unif}(G)}[\ell(\hat{f}(\varphi(g \cdot x)), f^*(x))]$ measures performance averaged over full orbits.*

When $\delta(\hat{f}) > 0$, the model exploits non-invariant features of the canonical form. Performance on canonical test data overestimates invariant prediction quality.

2.2 When Leakage Occurs

Proposition 1 (Informal)**.** *Canonicalization leakage occurs when two conditions hold: (i) non-invariant features of canonical representatives $\varphi_{non}(C(x))$ correlate with the target $f^*(x)$, and (ii) the learning algorithm can exploit this correlation.*

Condition (i) is subtle: even though f^* is G-invariant, the non-invariant features φ_{non} evaluated *at the canonical representative* carry target-correlated information, because the canonicalization operator C deterministically maps target values to specific regions of the non-invariant feature space. Condition (ii) holds for any sufficiently expressive learner.

2.3 Signal Taxonomy

We decompose the apparent performance of a model trained on canonical data into three components:

$$M_{\text{canon}} = S_{\text{inv}} + S_{\text{neural}} + L_{\text{canon}} \tag{1}$$

where S_{inv} is the performance from known G-invariant features (classical invariant ceiling), S_{neural} is additional invariant signal the network learns beyond classical measures, and L_{canon} is spurious canonicalization leakage. Each component is measured via CL-DIAG.

2.4 CL-DIAG: A Diagnostic Protocol

We propose a protocol for detecting and quantifying canonicalization leakage:

1. **Train on canonical data.** Train model $\hat{f}$ on $\mathcal{D}_C$; record canonical performance M_{canon}.
2. **Invariance test.** For a test subset, apply random $g \sim G$ transforms and measure prediction variance. Non-invariant predictions indicate leakage.
3. **Contamination decomposition.** Train separate models on each feature group φ_k; run the invariance test on each to identify which features carry canonical bias.
4. **Symmetry-consistent augmentation.** Augment training data with random G-transforms. This forces G-invariant representations, recovering invariant performance $S_{\text{inv}} + S_{\text{neural}}$.
5. **Causal control.** Compare group-consistent augmentation against matched-volume alternatives (data duplication, random noise) to isolate the symmetry-consistency mechanism.
6. **Signal decomposition.** Compute each component of Eq. (1): S_{inv} from invariant-feature-only models, $S_{\text{neural}} = \text{Perf}_G(\hat{f}_{\text{aug}}) - S_{\text{inv}}$, and $L_{\text{canon}} = M_{\text{canon}} - S_{\text{inv}} - S_{\text{neural}}$.

3 Boolean Circuit Complexity under NPN Symmetry

3.1 Circuit Complexity and NPN Equivalence

The *circuit complexity* $C(f)$ of a Boolean function $f : \{0,1\}^n \to \{0,1\}$ is the minimum number of binary gates (over the B_2 basis of all 16 two-input gates) required to compute f.

Two n-input Boolean functions are *NPN-equivalent* if one can be obtained from the other by permuting inputs (S_n), negating any subset of inputs ($\mathbb{Z}_2^n$), and negating the output ($\mathbb{Z}_2$). The NPN group $G = (S_n \ltimes \mathbb{Z}_2^n) \times \mathbb{Z}_2$ has order $|G| = n! \cdot 2^{n+1}$; for $n = 5$, $|G| = 7{,}680$. Circuit complexity is an NPN invariant.

Knuth computed exact minimum circuit sizes for all 616,126 NPN classes of 5-input functions [Knuth, 2015], and Goucher released the complete database [Goucher, 2019]. Each entry stores one canonical truth table per class, selected by lexicographic convention. The symmetry group is finite, the target is exact, the database is complete, and classical invariants are well-characterized.

Table 1: Feature groups and NPN invariance properties.

Group	Features	Dim	NPN-invariant?
A: Truth table bits	Raw $\{0, 1\}$ vector	32	No (bits rearrange and flip)
B: Fourier spectrum	Walsh-Hadamard coefficients	32	No (permute and sign-change)
C: Measures	7 invariant + 3 ordering-dependent	10	Partial (see text)

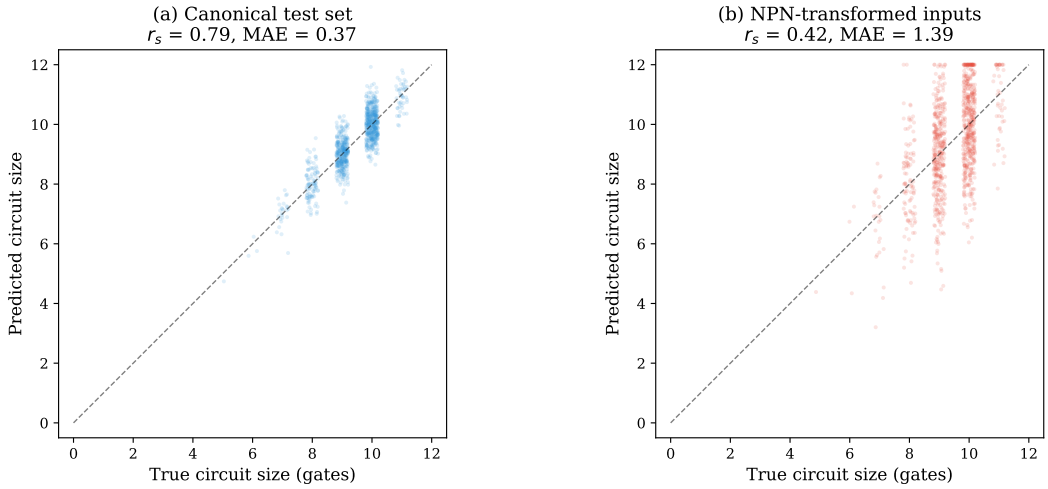

Figure 1: **Canonical vs. NPN-averaged predictions.** (a) Predictions on canonical test set show high correlation. (b) Same model on random NPN-equivalent inputs: correlation drops, with prediction std $\approx$1.2 gates across transforms and 0% consistency.

3.2 Feature Representations

We use three feature groups with distinct invariance properties (Table 1):

The full feature vector ($d = 74$) concatenates all three groups. The 7 NPN-invariant measures (Shannon entropy, spectral entropy, algebraic degree, nonlinearity, autocorrelation sum, sensitivity, total influence) form a leakage-free baseline: models trained only on these cannot exhibit leakage. The 3 ordering-dependent measures (LZ76, run length, gzip ratio) depend on bit ordering within the truth table. Architecture and training details are in Appendix A.

4 Experiments

We execute CL-DIAG end-to-end on the NPN5 circuit complexity task.

4.1 Baseline Performance on Canonical Representatives

An MLP trained on all 74 features achieves Spearman $r_s = 0.765$ on the canonical test set. Training separate models on individual feature groups (Table 2) shows that truth table bits ($r_s = 0.759$) and the Fourier spectrum (0.753) both exceed invariant measures (0.635).

4.2 NPN Invariance Test

An NPN-invariant complexity predictor must produce identical outputs for all elements of the same orbit. We test this by applying $K = 30$ random NPN transforms (input permutation, input negation, output negation) to 500 test functions and recording predictions under each transform.

The model is not NPN-invariant (Figure 1). Mean prediction standard deviation across transforms is 1.20 gates, and no test function achieves consistent rounded predictions across all transforms. On this 500-function evaluation subset, canonical Spearman is $r_s = 0.788$; NPN-averaged Spearman drops to 0.254. The model's predictions depend on properties specific to the canonical representation.

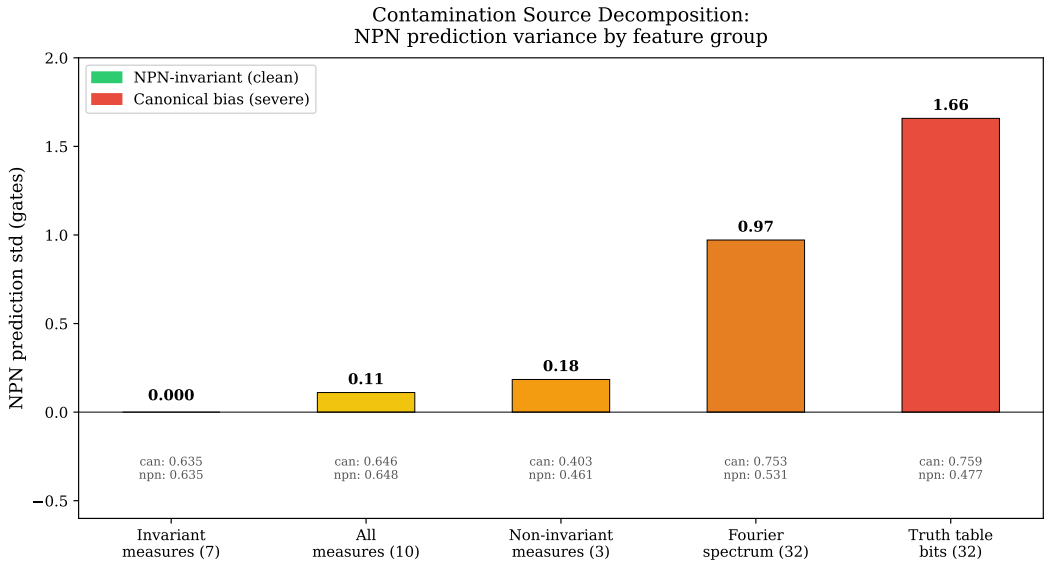

Figure 2: **Per-feature invariance testing.** NPN prediction std by feature group. Invariant measures show zero variation (std $= 0.000$); truth table bits are the most contaminated (std $= 1.658$). Values below bars: canonical and NPN-averaged r_s.

Table 2: Per-feature invariance testing. Each row: a model trained on one feature group.

Feature group	**Dim**	**Canon.** r_s	**NPN std**	**Consistency**	**NPN-avg** r_s
$C_{\text{invariant}}$ (7 measures)	7	0.635	**0.000**	**100%**	0.635
C_{all} (10 measures)	10	0.646	0.111	51%	0.648
$C_{\text{non-inv}}$ (3 measures)	3	0.403	0.184	2.4%	0.461
B (Fourier spectrum)	32	0.753	0.972	0%	0.531
A (truth table bits)	32	0.759	1.658	0%	0.477

This recontextualizes the feature ablation from Section 4.1: "bits $\approx$ Fourier" does not reflect learned spectral analysis. Both feature groups achieve similar canonical performance because both exploit canonical-form artifacts, and both fail under NPN transforms.

4.3 Contamination Source Decomposition

We train separate models on each feature group and run the invariance test on each (Figure 2, Table 2).

The 7 NPN-invariant measures show zero leakage: std $= 0.000$, consistency $= 100\%$, canonical r_s equals NPN-averaged $r_s = 0.635$. This establishes the **classical invariant ceiling**: the best performance from known invariant features. Truth table bits are the most contaminated (std $= 1.658$), and the Fourier spectrum is also contaminated (std $= 0.972$). Both show 0% consistency.

Adding the 3 ordering-dependent measures to the invariant set raises canonical r_s from 0.635 to 0.646 but drops consistency from 100% to 51%.

4.4 Signal Redistribution under Symmetry Exposure

We append k randomly NPN-transformed copies of each training sample ($k \in \{0, 1, 3, 7\}$), recompute features from each transformed truth table, and retrain. Results appear in Table 3 and Figure 3.

Canonical r_s stays flat ($\sim$0.77–0.79) while invariant performance rises, suggesting that the total predictive signal is approximately constant and augmentation progressively transfers it from leakage to invariant structure. All invariance metrics improve monotonically: NPN-averaged r_s rises from 0.254 to 0.777, prediction std drops from 1.196 to 0.091, consistency increases from 0% to 65.6%.

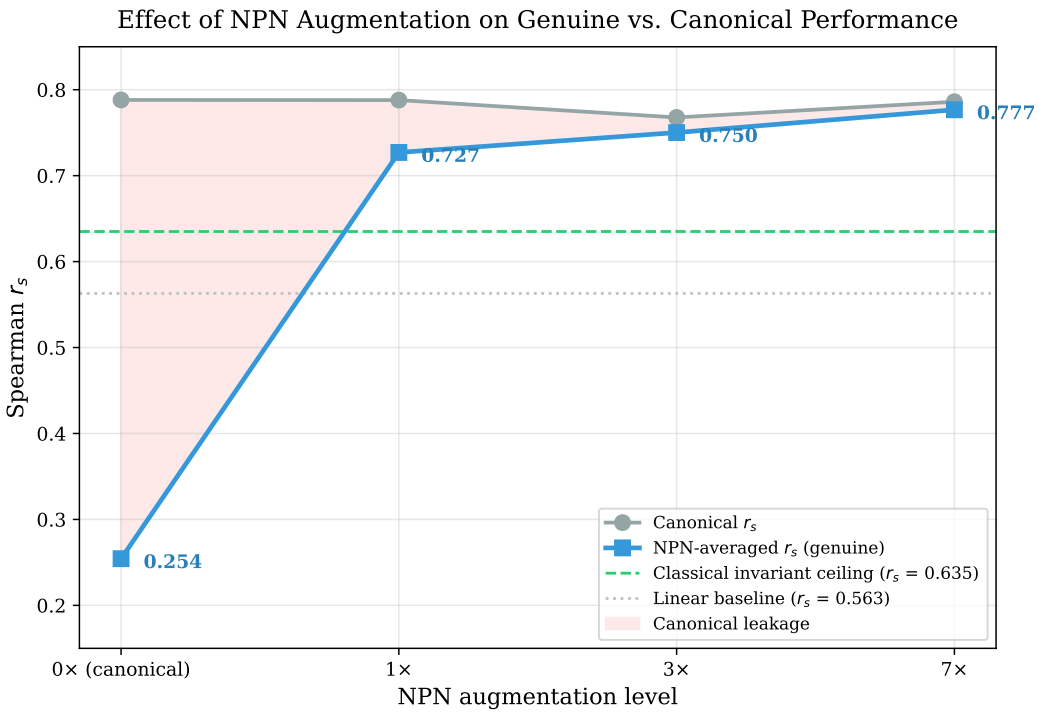

Figure 3: **Signal redistribution under symmetry exposure.** As augmentation level k increases, predictive signal transfers from leakage L_{canon} (shaded gap) to invariant structure S_{neural} (blue curve). The canonical r_s (grey) stays flat while NPN-averaged r_s rises monotonically—augmentation redistributes signal from leakage to invariant structure without changing total canonical performance.

Table 3: Signal redistribution under increasing symmetry exposure (K=30 transforms, 500 test functions).

Config	**Train size**	**NPN std**	**Consistency**	**Canon.** r_s	**NPN-avg** r_s
0× (baseline)	431K	1.196	0.0%	0.788	0.254
1×	862K	0.147	37.8%	0.788	0.727
3×	1.73M	0.115	53.0%	0.768	0.750
7×	3.45M	**0.091**	**65.6%**	0.786	**0.777**

The 7× model reaches $r_s = 0.777$, 14pp above the classical invariant ceiling of 0.635, indicating NPN-invariant structure not captured by the 7 classical measures.

4.5 Causal Identification: Symmetry vs. Regularization

We isolate the active mechanism. Three matched-volume conditions (all at 2× training volume, 862K samples) test whether the invariance gain is attributable to group-consistent transformation specifically, or to data volume and regularization generically:

1. **NPN augmentation**: canonical + 1 NPN-transformed copy (group-consistent)
2. **Data duplication**: canonical repeated 2× (same volume, zero new information)
3. **Noise augmentation**: canonical + copy with Gaussian noise (σ=0.1) on features

The three conditions share identical training volume but differ in what structure the additional data carries. NPN augmentation adds orbit samples that are group-consistent; noise adds entropy that is group-unrelated; duplication adds nothing. If volume or regularization were sufficient, duplication and noise should match NPN augmentation. They do not (Table 4).

Table 4: Matched-volume control (separate training run). All conditions at $2\times$ volume (862K samples).

Condition	**Val r_s**	**NPN std**	**Consistency**	**NPN-avg r_s**
NPN augmentation	0.761	**0.141**	**38.8%**	**0.728**
Data duplication	0.787	1.600	0.0%	0.172
Noise augmentation	0.779	0.398	0.0%	0.691

Table 5: Signal decomposition of canonical performance on the 500-function evaluation subset (Eq. (1)).

Component	**Source**	**r_s range**	Δ
S_{inv}: Classical invariant signal	7 invariant measures (MLP)	$0.563 \to 0.635$	$+0.072$
S_{neural}: Neural invariant signal	Learned from augmented bits/Fourier	$0.635 \to 0.777$	$+0.142$
L_{canon}: Canonicalization leakage	Canonical-form artifacts	$0.777 \to 0.788$	$+0.011$
M_{canon}: Total canonical performance			0.788

Data duplication achieves the highest validation r_s (0.787) but the worst NPN-averaged r_s (0.172)—worse than the unaugmented baseline (0.254). With identical copies, the model reinforces its reliance on canonical artifacts. **Noise augmentation** partially disrupts canonical patterns (NPN std $= 0.40$ vs. 1.60 for duplication) and recovers $r_s = 0.691$, but consistency remains 0%. Only NPN augmentation yields nonzero consistency (38.8%); it also achieves the highest orbit-averaged r_s.

4.6 Signal Decomposition

After symmetry-consistent augmentation, the corrected attribution of canonical performance (Eq. (1)):

The corrected leakage of 1.1pp is negligible—but only because augmentation has already transferred signal to S_{neural}. Before diagnosis, the same model attributes $r_s = 0.788$ to the canonical form, of which only 0.254 survives orbit averaging. A linear baseline on the same features achieves $r_s = 0.563$; the nonlinear MLP on invariant measures adds $+7.2$pp to reach $S_{\text{inv}} = 0.635$. Symmetry-augmented training reveals $+14.2$pp of neural invariant signal (S_{neural}).

5 Related Work

Canonical forms in molecular ML. Bjerrum [2017] showed that randomized SMILES [Weininger, 1988] improve molecular property prediction over canonical SMILES, confirmed by Arús-Pous et al. [2019] for generative models and motivating alternative representations such as SELFIES [Krenn et al., 2020]. Steshin [2025] tested chemical LLM robustness to SMILES variation. These studies documented the phenomenon empirically in a single domain. The framework in Section 2 provides a group-theoretic account; CL-DIAG adds per-feature invariance testing and matched-volume controls (Section 4.5) that were absent from the original studies.

Equivariant architectures and canonicalization. The geometric deep learning program [Bronstein et al., 2021] provides a framework for building G-invariant models. Equivariant networks [Cohen and Welling, 2016, Schütt et al., 2018, Satorras et al., 2021, Maron et al., 2019] can achieve G-invariance by construction, eliminating leakage architecturally. Frame averaging [Puny et al., 2022] achieves invariance by averaging over frames; learned canonicalization [Kaba et al., 2023] trains a canonicalization function end-to-end. Dym et al. [2024] proved that continuous canonicalization is generally impossible, and Ma et al. [2024] connected canonicalization to frame averaging. These works treat canonicalization as a *solution*; we identify it as a potential *data-level problem* for non-equivariant models on pre-canonicalized data—a setting that remains common in practice [Xu et al., 2019].

Group-theoretic augmentation. Chen et al. [2020] showed that augmentation amounts to orbit averaging with variance-reduction guarantees. Dao et al. [2019] analyzed augmentation through a kernel-theoretic lens, showing that it implicitly changes the function class. Our matched-volume

control (Section 4.5) additionally tests whether the gain is attributable to group-consistency or to volume alone, by comparing orbit augmentation against duplication and noise at equal volume.

Boolean function ML. Ranga et al. [2025] predicted Boolean nonlinearity with neural networks; representation invariance under the symmetry group was not part of the evaluation protocol. The invariance test from CL-DIAG (Section 2.4) provides a way to add such evaluation.

Shortcut learning and spurious correlations. Canonicalization leakage is an instance of shortcut learning [Geirhos et al., 2020] arising from the interaction between group symmetry and canonical preprocessing. It is related to broader work on spurious correlations and distribution shift [Beery et al., 2018, Sagawa et al., 2020], but with a distinctive structural property: the shortcut is induced by the canonicalization procedure, not a contingent property of a particular dataset. This makes it simultaneously more predictable (it occurs whenever conditions (i)–(ii) of our informal proposition hold) and harder to detect (canonical test sets exhibit the same bias as training sets, so standard train/test evaluation cannot reveal the failure).

6 Discussion

A representation-dynamics view. The monotonic augmentation curve (Table 3) supports a structural interpretation: canonicalization leakage induces a systematic mismatch between the learned predictor and the invariant target; symmetry-consistent augmentation acts as a progressive projection that transfers predictive signal from non-invariant artifacts (L_{canon}) to invariant structure (S_{neural}), while leaving S_{inv} unchanged. This complements the orbit-averaging framework of Chen et al. [2020], which focuses on variance reduction rather than spurious signal.

Applicability beyond circuit complexity. CL-DIAG applies whenever canonical representatives are used with known group symmetry. Candidate domains include molecular property prediction (canonical SMILES), graph classification (nauty labeling), and crystallography (Niggli reduction). We do not claim leakage exists in all cases—equivariant architectures may be immune by construction—but the protocol provides a diagnostic test. NPN5 was chosen because it provides the conditions for controlled identification: a complete database, exactly known targets, and a fully characterized finite group.

Limitations. Our theoretical analysis is informal; formal bounds on canonicalization leakage remain open. We test only MLP architectures; equivariant architectures may resist leakage by construction. Symmetry-consistent augmentation requires sampling G-transforms, which is expensive for large or continuous groups. Our case study covers $n = 5$ only; generalization to larger Boolean functions is not tested. The signal taxonomy assumes approximate additivity; S_{neural} and L_{canon} may interact in practice.

7 Conclusion

We introduced *canonicalization leakage*—a failure mode where models trained on canonical representatives under group symmetry exploit representation artifacts rather than learning invariant structure. CL-DIAG detects and quantifies this failure through invariance testing, contamination decomposition, symmetry-consistent augmentation, and matched-volume controls. On Boolean circuit complexity prediction, standard evaluation on canonical data inflates apparent invariant performance ($r_s = 0.788$ canonical vs. 0.254 orbit-averaged). Symmetry-consistent augmentation recovers $r_s = 0.777$, 14pp above the classical invariant ceiling (0.635), and matched-volume controls isolate group-consistent transformation from regularization as the source of the gain.

The framework in Section 2 offers a group-theoretic account of earlier observations of SMILES sensitivity in molecular ML. Invariance testing requires only random group transforms and can be added to existing evaluation protocols for symmetry-reduced datasets.

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NeurIPS Paper Checklist

1. **Claims**

   Answer: [Yes]

   Justification: The abstract and introduction clearly state five contributions, each supported by experimental evidence in Section 4. Claims are scoped to the NPN5 case study with discussion of broader applicability in Section 6.

2. **Limitations**

   Answer: [Yes]

   Justification: Section 6 discusses limitations including informal theoretical analysis, MLP-only architecture, computational cost of augmentation for large groups, n=5 only, and approximate additivity of signal taxonomy.

3. **Theory assumptions and proofs**

   Answer: [N/A]

   Justification: Proposition 1 is explicitly labeled as informal. The paper is primarily empirical/methodological. Rigorous PAC-style bounds are noted as future work.

4. **Experimental result reproducibility**

   Answer: [Yes]

   Justification: Section 3.2 and Appendix A specify all hyperparameters, data splits, and training details. The dataset is publicly available (MIT license). Random seeds are fixed for reproducibility.

5. **Open access to data and code**

   Answer: [Yes]

   Justification: Code and data will be released upon acceptance. The dataset (Goucher's optimal5) is already publicly available under MIT license.

6. **Experimental setting/details**

   Answer: [Yes]

   Justification: Section 3.2 and Appendix A provide architecture, optimizer, learning rate, batch size, epochs, data split, and feature computation details.

7. **Experiment statistical significance**

   Answer: [No]

   Justification: We report single-run results as the dataset is complete (all 616,126 NPN classes) rather than sampled, and the NPN invariance test aggregates over 500 functions $\times$ 30 transforms, providing sufficient statistical power. The key finding (0% consistency) is deterministic, not stochastic.

8. **Experiments compute resources**

   Answer: [Yes]

   Justification: Appendix A reports hardware (i7-12700K + RTX 3090) and total compute ($<$ 4 GPU-hours).

9. **Code of ethics**

   Answer: [Yes]

   Justification: This research involves mathematical objects (Boolean functions) with no human subjects, privacy, or fairness concerns.

10. **Broader impacts**

    Answer: [N/A]

    Justification: This is foundational methodology research on representation bias in symmetry-reduced datasets. It has no direct negative societal applications. Positive impact: improving reliability of ML evaluations.

11. **Safeguards**

Answer: [N/A]

Justification: No models or datasets with risk of misuse are released.

12. **Licenses for existing assets**

    Answer: [Yes]

    Justification: The optimal5 dataset is released under MIT license by Goucher (2019). PyTorch and NumPy are BSD-licensed.

13. **New assets**

    Answer: [Yes]

    Justification: Code implementing CL-DIAG and the diagnostic protocol will be released upon acceptance under an open-source license.

14. **Crowdsourcing and research with human subjects**

    Answer: [N/A]

    Justification: No human subjects involved.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Answer: [N/A]

    Justification: No human subjects involved.

16. **Declaration of LLM usage**

    Answer: [Yes]

    Justification: LLMs were used for writing assistance (drafting, editing, and proofreading). They are not a component of the core methodology or experimental pipeline.

A Experimental Details

Model and training. 3-layer MLP (256-256-128) with BatchNorm, GELU, Dropout (0.1). AdamW optimizer [Loshchilov and Hutter, 2019] (lr $= 10^{-3}$, weight decay 10^{-4}), cosine annealing, 60 epochs, batch size 2048. All experiments on a single NVIDIA RTX 3090. Random seed fixed across all runs for reproducibility.

Dataset. The NPN5 database contains 616,126 entries: 32-bit canonical truth tables with optimal circuit structures. Circuit sizes range 0–12 gates (mode at size 10, $\sim$48%). Split: 431,288 train (70%), 92,419 validation (15%), 92,419 test (15%), stratified by circuit size.

Feature computation. Truth table bits: 32-bit truth table as $\{0, 1\}$ vector. Fourier: Walsh-Hadamard transform of $\{-1, +1\}$ encoding, normalized by 2^n. Measures: Shannon entropy, spectral entropy, LZ76, run length, gzip ratio, algebraic degree, nonlinearity, autocorrelation sum, sensitivity, total influence. Z-normalized using training statistics.

NPN transforms. Each transform: (1) random $\sigma \in S_5$, (2) random negation mask $\eta \in \{0, 1\}^5$, (3) random output negation $\nu \in \{0, 1\}$. Transformed table: $T'[\text{idx}] = T[\sigma^{-1}(\text{idx} \oplus \eta)] \oplus \nu$.

Compute. Single workstation (i7-12700K + RTX 3090 24GB). $0\times$ baseline: $\sim$4 min; $7\times$ augmentation: $\sim$29 min including feature computation. Total: < 4 GPU-hours.

B Additional Results

Architecture ablation. A 4-layer MLP (512-512-256-128, 468K params) achieves $r_s = 0.790$ vs. 0.765 for the standard 3-layer. A wider 3-layer (512-512-512, 567K params) reaches $r_s = 0.791$ but with worse MAE. Both are equally susceptible to canonicalization leakage (Section 4.2).

Classification. A 13-class classification model achieves 70.0% exact match and MAE $= 0.304$, compared to 67.6% and 0.405 for regression. Both are equally susceptible to canonicalization leakage.